SmartCare

Hospital Management System

Software Requirements Specification (SRS)

Integrated & Consolidated — Version 1.0

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Faculty: Faculty of Engineering
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1. Introduction

1.1 Purpose

This document defines the consolidated Software Requirements Specification (SRS) for SmartCare, a web-based Hospital Management System. It serves as the authoritative reference for developers, testers, and stakeholders involved in design, implementation, and validation.

1.2 Project Overview

SmartCare is a comprehensive, web-based platform designed to streamline healthcare operations by centralizing appointments, medical records, prescriptions, and billing into a single accessible interface. It serves three primary user roles — Administrators, Doctors, and Patients — and provides analytics dashboards to help hospital staff monitor and improve service quality.

1.3 Scope

The system covers the following functional areas:

- User account management and authentication
- Appointment scheduling and management
- Patient medical records and history
- Prescription management
- Billing and payment processing
- Notification and alert system
- Analytics and reporting dashboard

1.4 Definitions and Acronyms

Term / Acronym	Definition
HMS / SRS	Hospital Management System / Software Requirements Specification
EHR / EMR	Electronic Health Record / Electronic Medical Record
Admin	Hospital Administrator — manages system-level operations
HTTPS/TLS	Secure communication protocol used between client and server
UC / FR / NFR	Use Case / Functional Requirement / Non-Functional Requirement
HIPAA	Health Insurance Portability and Accountability Act (or equivalent local regulation)

2. Stakeholders and User Roles

SmartCare involves the following stakeholders:

Role	Description	Key Responsibilities
Administrator	Hospital staff with full system access	Manage users, view analytics, configure settings, manage billing
Doctor	Medical professional registered in the system	View/update patient records, manage appointments, issue prescriptions
Patient	Registered hospital patient	Book appointments, view records/prescriptions, make payments, receive notifications
System	The SmartCare platform (automated actor)	Send notifications, generate invoices, enforce access control, run backups

3. Functional Requirements

Requirements have been consolidated and minimized from both source documents, eliminating duplication while retaining full functional coverage.

3.1 Authentication & Account Management

ID	Actor	Requirement	Use Case
FR-01	All Users	Users shall register with name, email, password, and role (Patient/Doctor).	UC-01
FR-02	All Users	Users shall log in using their registered email and password. Upon successful authentication, the system shall redirect each user to their role-specific dashboard (Admin, Doctor, or Patient).	UC-02
FR-03	All Users	Users shall be able to reset their password via email verification.	UC-03
FR-04	Admin	Administrators shall create, edit, activate, or deactivate user accounts and manage inactive session timeouts.	UC-04

3.2 Appointment Management

ID	Actor	Requirement	Use Case
FR-05	Patient	Patients shall search for doctors by name, specialty, or date, then book, reschedule, or cancel appointments.	UC-05
FR-06	Doctor / Admin	Doctors shall view their daily and weekly schedule and confirm, reschedule, or cancel appointments. Admins shall oversee all hospital appointments.	UC-06
FR-07	System	The system shall send automated reminders to patients 24 hours before appointments.	UC-05, UC-06

3.3 Medical Records Management

ID	Actor	Requirement	Use Case
FR-08	Doctor	Doctors shall create, update, and upload documents or diagnostic reports to a patient's medical record after each visit.	UC-08
FR-09	Doctor / Patient	Doctors shall view a patient's full medical history. Patients shall view their own records and lab reports. Records shall be accessible only to authorized users.	UC-07

3.4 Prescription Management

ID	Actor	Requirement	Use Case
FR-10	Doctor	Doctors shall create digital prescriptions (drug name, dosage, frequency, duration) linked to a patient visit.	UC-09
FR-11	Patient / System	Patients shall view and download active/past prescriptions. The system shall notify patients when a new prescription is issued.	UC-10

3.5 Billing & Payment

ID	Actor	Requirement	Use Case
FR-12	System / Admin	The system shall automatically generate an invoice upon appointment completion. Admins shall manage billing records and process refunds.	UC-11, UC-13
FR-13	Patient	Patients shall view billing history, pay pending invoices via card or digital wallet, and receive a PDF receipt upon payment.	UC-12

3.6 Notifications

ID	Actor	Requirement	Use Case
FR-14	System	The system shall send email and in-app notifications for appointments, prescriptions, and billing events. Admins shall receive alerts for critical system errors.	UC-14
FR-15	All Users	Users shall be able to configure their notification preferences.	UC-14

3.7 Analytics & Reporting

ID	Actor	Requirement	Use Case
FR-16	Admin	Administrators shall access a real-time dashboard showing KPIs: total appointments, revenue, active patients, and doctor utilization.	UC-15
FR-17	Admin	The system shall generate reports on doctor performance, appointment trends, and billing summaries, exportable in PDF and Excel formats.	UC-15

4. Non-Functional Requirements

ID	Category	Requirement	Priority
NFR-01	Performance	Any page shall load within 3 seconds under normal conditions; appointment booking shall complete within 2 seconds of confirmation.	High
NFR-02	Performance	The system shall support at least 500 concurrent users without performance degradation.	High
NFR-03	Security	Passwords shall be stored using a strong hashing algorithm (e.g., bcrypt). All client-server communication shall be encrypted via HTTPS/TLS.	High
NFR-04	Security	Role-based access control shall prevent unauthorized data access. Doctors may only access records of patients assigned to them.	High
NFR-05	Availability	The system shall maintain 99.5% uptime. Planned maintenance shall not exceed 4 hours per month, scheduled outside clinic hours.	High
NFR-06	Usability	The interface shall be responsive across desktop, tablet, and mobile. New users shall complete core tasks within 5 minutes without prior training.	Medium
NFR-07	Scalability	The architecture shall support horizontal scaling to accommodate hospital growth.	Medium
NFR-08	Maintainability	The codebase shall follow modular design principles to facilitate future enhancements.	Medium
NFR-09	Compliance	The system shall comply with applicable healthcare data privacy regulations (e.g., HIPAA or equivalent).	High
NFR-10	Reliability	The system shall perform automated daily backups of all patient data.	High

5. System Architecture

SmartCare follows a three-tier Client-Server Architecture:

Layer	Description
Presentation Layer	Web browser interface (Chrome, Firefox, Edge, Safari) with distinct, responsive dashboards for each user role (Patient, Doctor, Admin).
Application Layer	Server-side business logic including appointment scheduling, billing calculations, notification triggers, and access control enforcement.
Data Layer	Centralized relational database (MySQL or PostgreSQL) storing user credentials, EHRs, appointments, billing records, and operational data.

6. Use Case Overview

6.1 Use Case Summary

UC ID	Use Case	Actor(s)	Related FRs
UC-01	Register Account	Patient, Doctor	FR-01
UC-02	Login to System	All Users	FR-02
UC-03	Reset Password	All Users	FR-03
UC-04	Manage User Accounts	Administrator	FR-04
UC-05	Book Appointment	Patient	FR-05, FR-07
UC-06	Manage Appointment	Doctor, Admin	FR-06, FR-07
UC-07	View Medical Records	Doctor, Patient	FR-09
UC-08	Update Medical Records	Doctor	FR-08
UC-09	Issue Prescription	Doctor	FR-10, FR-11
UC-10	View Prescription	Patient	FR-11
UC-11	Generate Invoice	System	FR-12
UC-12	Process Payment	Patient	FR-13
UC-13	Manage Billing Records	Administrator	FR-12
UC-14	Send Notification	System	FR-14, FR-15
UC-15	View Analytics Dashboard	Administrator	FR-16, FR-17

7. System Constraints and Assumptions

7.1 Constraints

- The system must be web-based and accessible via modern browsers (Chrome, Firefox, Edge, Safari).
- All client-server communication must use HTTPS/TLS encryption.
- Patient data must be stored securely in compliance with applicable data protection regulations.
- The system must remain functional during partial server failures via graceful degradation.

7.2 Assumptions

- All users have internet access and a modern web browser.
- Hospital billing rates and doctor schedules are configured by the administrator before system use.
- An email service is available for notifications and password-reset links.
- A payment gateway integration is available for online transactions.

8. Time Plan

Phase	Activities	Timeline
Phase 1: Requirements & Analysis	Gather requirements, define stakeholders, review SRS	Weeks 1–2
Phase 2: System Design & UML Modeling	Architecture design, UML diagrams (use case, class, sequence)	Weeks 3–4
Phase 3: Implementation / Prototyping	Frontend & backend development, module integration	Weeks 5–8
Phase 4: Testing & Validation	Unit testing, system testing, bug fixes	Weeks 9–10
Phase 5: Documentation & Presentation	Final documentation, demo preparation, submission	Weeks 11–12

9. Requirements Traceability Matrix

The following matrix maps consolidated Functional Requirements to their corresponding Use Cases.

FR ID	Summary	Use Case(s)
FR-01	User Registration	UC-01
FR-02	Authentication & Role-Based Access Control	UC-02
FR-03	Password Reset	UC-03
FR-04	Admin User & Session Management	UC-04
FR-05	Search Doctors & Book / Cancel Appointments	UC-05
FR-06	Doctor / Admin Appointment Management	UC-06
FR-07	Automated Appointment Reminders	UC-05, UC-06
FR-08	Create & Update Patient Medical Records	UC-08
FR-09	View Medical Records (authorized access)	UC-07
FR-10	Issue Digital Prescriptions	UC-09
FR-11	View / Download Prescriptions & Notifications	UC-10
FR-12	Auto Invoice Generation & Admin Billing Mgmt.	UC-11, UC-13
FR-13	Patient Payment & PDF Receipts	UC-12
FR-14	Notification System (all events)	UC-14
FR-15	User Notification Preferences	UC-14
FR-16	Analytics Dashboard (KPIs)	UC-15
FR-17	Reporting & Export (PDF / Excel)	UC-15

10. References

- Sommerville, I. (2020). Engineering Software Products — An Introduction to Modern Software Engineering. Pearson.
- Course Lectures: Software Engineering, Dr. Samar A. Said (2025/2026), Helwan University.
- SmartCare Project Report V1 — Team 2, Helwan University (April 2026).
- SmartCare SRS & Use Cases Document — Team 2, Helwan University (April 2026).